

FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

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● LIVE FROM PLAYHQ



USER GUIDE

Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

5 s
The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

31
Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17
Competition logo animations, and six intro and outro styles.

1
Operator on a followed night. Sequences run the run-up, the breaks and full time.

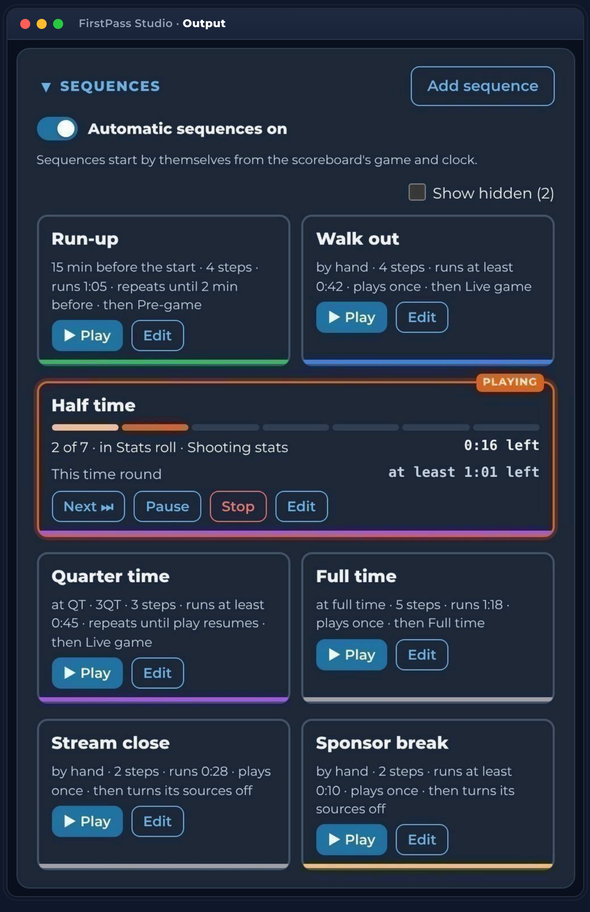
IN
PlayHQ
The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC
FirstPass Studio
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT
Your switcher
A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.





The Output page runs the show; the super shot plays by itself.

INSIDE

01 **What Studio is** — The app, the words used and the pages you work in.

02 **Getting started** — Install, on air and off air, the 1P menu, the checklist.

03 **Game night, the automated way** — Follow PlayHQ, the base scene, automatic layers, sequences, music, output.

04 **Running it by hand** — Manual scoring and clock, and a two-person crew.

05 **The graphics** — Every graphic, pictured, with the options worth knowing.

06 **Competitions setup** — PlayHQ link, rules, finals, colours, logos, live viewers.

07 **Sponsors** — Tiers, spots, ads, sponsored moments and proof of play.

08 **Athletes** — Photos, silhouettes, positions and games played.

09 **Stream Deck** — The plugin's keys, the links and timed keys.

10 **Settings and care** — Data folder, iCloud, backups, updates and fixes.

11 **Quick reference** — Keyboard, the night in nine lines, where things are set.

The sample league — Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.

CHAPTER 1 OF 11

What Studio is

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio

The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air

Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page

One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source

A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene

Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)

A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence

A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition

Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

The data folder

Where Studio saves everything, outside the app, so updates never touch it.

1.2 The pages you work in

Every Studio window has the same bar at the top, with the FirstPass Studio mark and the tabs in three groups by what you are doing. **Run** is the night itself: **Output**, where the show is built and sent out, and **Scoreboard control**, which runs the board. **Plan** is before the night: the **Production Planner** plans the court and the commentators, and **Socials** makes the night's pictures for social media. **Set up** is done once: **Recordings**, **Competitions setup**, **Sponsors**, **Athletes** and **Devices**. **Docs** sits on its own at the right: it is this guide, with the bar on top. In the app, ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices and Docs.

The right of the bar carries the status chips, on every page, so you can see the night from wherever you are: **ON AIR** in orange like a tally light, with a dot per output that is solid in the output's colour while it sends and hollow while it doesn't (or **Off air**); the game on the board, its dot orange before the clock has started, green from the first clock start or score until an hour after the game ends, then grey; PlayHQ, whose dot is the link alone (green when PlayHQ answers, amber when a reading is overdue or reads **your score held** while PlayHQ is behind a hand score, red when it keeps failing, purple and **PlayHQ not live** when PlayHQ answers but the game isn't being scored live, grey when scores are by hand) and whose words give the quarter and clock, or **game completed** once PlayHQ closes the game (**full time** when the last quarter is run down by hand); the sequence playing with its step and the time left on it, or the triggered sequence counting down to start; **Run order** ▶ with what Play next plays and ⏭ with what follows it (also what Skip leaves at the head), shown whatever is playing and gone when the run order is empty; **ATEM** while a switcher is set up (green when it answers, red when it doesn't), and **REC** while a game records (grey while Record games waits for one). The chips are status, not links; the tabs take you to a page.

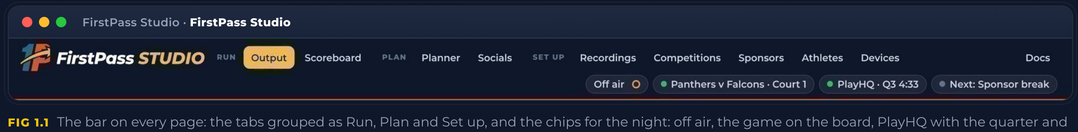
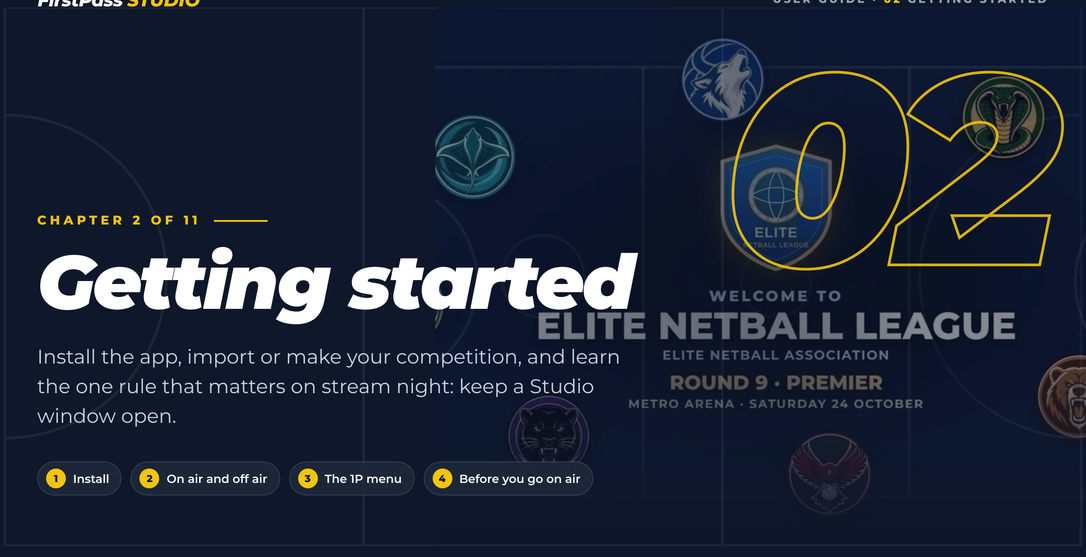


FIG 1.1 The bar on every page: the tabs grouped as Run, Plan and Set up, and the chips for the night: off air, the game on the board, PlayHQ with the quarter and clock, and what the run order plays next.

> ***Studio does one job on stream night: graphics.***
>
> CHAPTER 1 · WHAT STUDIO IS

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.



CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01
Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02
The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03
Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04
Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

The pages zoom like Safari's, for a small screen: **Zoom In** (⌘+), **Zoom Out** (⌘−) and **Actual Size** (⌘⌥0) in the View menu, which shows the level, or a pinch on the trackpad. Every Studio window zooms together and the level is kept on that Mac; while it isn't 100% a chip at the right of the bar shows it, and a click on the chip puts it back. The Docs tab, this guide, has its contents at the left on a wide window.

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The bar's chips show the game on the board, whether PlayHQ is answering (amber when overdue, red after a minute of errors, purple when the game isn't being scored live) and which outputs are on air, on every page; the Output page's status board shows the score and the rest of the night. If anything is wrong, the page you keep open says so.

2.3 The 1P menu

The 1P icon in the menu bar is Studio's home: it opens Studio, its pages and this guide, backs the settings up and restores them, and has **Open at login**, **Appearance** (the Studio pages follow the Mac's light or dark look unless pinned here; the graphics never change) and **Check for updates**. Chapter 10 lists every item. Updates download in the background and install when the app quits, never mid-stream.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> ***The one rule that matters on stream night: keep a Studio window open.***

CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01
Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02
Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link and the scoring window, and that the Production Planner has tonight's court.

03
Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04
Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05
Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06
Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07
Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08
Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

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Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

1 Two people, not one · 2 Switching to manual · 3 The score · 4 The clock and the quarter · 5 Breaks and celebrations by hand · 6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> **If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.**
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again. If PlayHQ is still behind your score when you press it (the scorebench hasn't caught up yet), the board keeps your score, quarter and clock until PlayHQ catches up or has something new from the scorer, then follows it; the PlayHQ chip reads **your score held** meanwhile. So you can press Follow PlayHQ as soon as the scorebench is back without losing the goals you scored by hand.

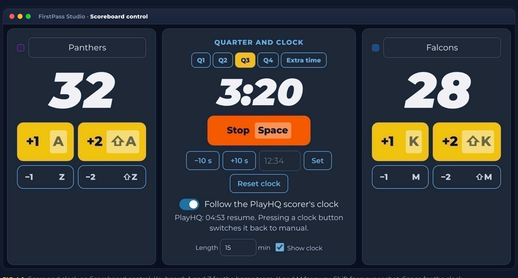


FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away. Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

If the scorebench drops off line instead (its wifi, say), PlayHQ still answers but its score stops moving. Studio spots this two ways: PlayHQ says the game is under way while its live feed says it isn't being scored (or it's five minutes past the start), or nothing new comes from the scorer for four minutes while its clock runs. Then the bar's PlayHQ chip turns purple and reads **PlayHQ not live**, and the status board's PlayHQ line reads **Not live**; hover over either for why. Fix the scorebench if you can, or score by hand until it's back and press Follow PlayHQ: your score stays on the board until PlayHQ catches up. When PlayHQ catches up with several goals at once, the score jumps straight to it, the goals that came together show as a gap on the last 10 goals and the score worm (their order and times aren't known), and no super shot or moment plays for them. If it catches up before the next quarter starts, as it does when the scorebench comes back at quarter time, that quarter's score still counts on the game stats.

4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length: a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.

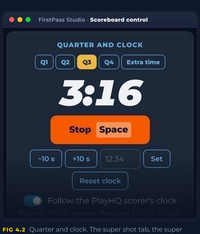


FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays:** home or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals



FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 5 OF 11

The graphics



Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its add-on. Many options named here are the ones worth knowing.



All of them follow the same rules: genre and court picked on Scoreboard control, take me competitors column, font and type, from its setup, and fit long team names. Cards in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its title, its options and a Preview button, is also listed under **Utilities** on the Devices tab.

5.1 Always on



Scoreboard. Score, time, and sets in one bar, twice as wide when a graphic with the competitors and set lines is up. Fill the name in the middle, to exactly as wide as both yellows, at both yellows. Scoreboard control, Competitors tab, Name. Shorten it to fit. Turn off Show sets to hide the set line. Turn off Show time to hide the clock. Pick the colour from the Theme tab.

Score Ticker. The scores from tonight's other games scrolling along the bottom, leaving the competitors' scores. Scoreboard control, Competitors tab, Ticker. Separate them with a comma or a semicolon. Send new a finished game's scores. Send new a finished game's scores. Send new a finished game's scores

CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

PlayHQ · Names on the graphics · Game rules · Finals · Game night · Colours and font · Logos · Live viewers · Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.

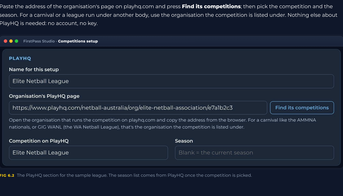


FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMHNA Nationals, GIG WANs, and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.

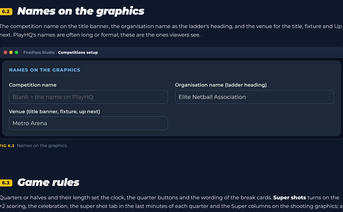


FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.

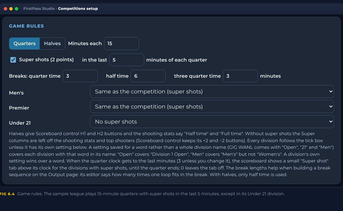
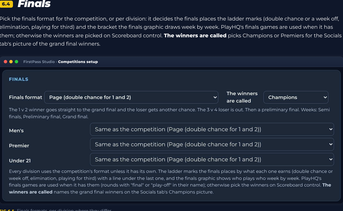


FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.

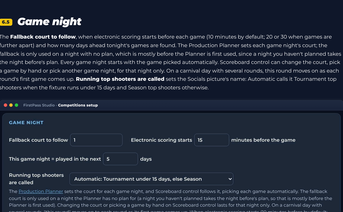


Halves give Scoreboard control HI and HZ buttons and the shooting stats say "Half time" and "Full time". Without super shots the Super columns are left off the shooting stats and top shooters (Scoreboard control keeps +2 and -2 buttons). Every division follows the tick box unless it has its own setting below. A setting saved for a word rather than a whole division name (GIG WANs, names with "Open", "2" and "Men") covers each division with that word in its name. "Open" covers "Division 1 Open". "Men" covers "Men's" but not "Womens". A division's own setting wins over a word. When the carnival clock gets to the last minutes (3 unless you change it), the scoreboard shows a small "Super shot" tab above its clock for the divisions with super shots, until the quarter ends. It leaves the tab off. The break lengths help when building a break sequence on the Output page: its editor says how many times one loop fits in the break. With halves, only half-time is used.

FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control. **The winners are called** picks Champions or Premiers for the Social tab's picture of the grand final winners.



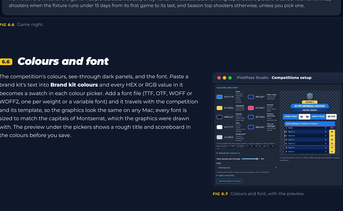
The 1-v-2 winner goes straight to the grand final and the loser gets another chance. The 3-v-4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals games are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on Scoreboard control. **The winners are called** names the grand final winners on the Socials tab's Champions picture.

FIG 6.5 Finals formats, per division where they differ.

6.5 Game night

The **Fallback court to follow**, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. The Production Planner sets each game night's court; the fallback is only used on a night with no plan, which is mostly before the Planner is first used, since a night you haven't planned takes the night before's court. Every grand final is set aside automatically, Scoreboard control can change the court, pick a game by hand or pick another game night, for that night only. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.



The Production Planner sets the court for each game night, and Scoreboard control follows it, picking each game automatically. The fallback court is set in case a night the Planner has no plan for (a night you haven't planned takes the night before's plan, so that is mostly before the Planner is first used). Changing the court or picking a game by hand on Scoreboard control lasts for that night only. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand. The Socials tab's running top shooters picture is called Tournament top shooters when the fixture runs under 15 days from its first game to its last, and Season top shooters otherwise, unless you pick one.

FIG 6.6 Game night.

6.6 Colours and font

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one file's weight or a variable font) and it teases with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.

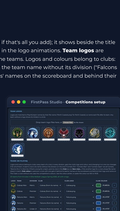


FIG 6.7 Colours and font, with the preview.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the banners and logos, above the title with its option on, in the corner of the ladder and in the logo animation. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 2U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash, without one the competition's colours are used.

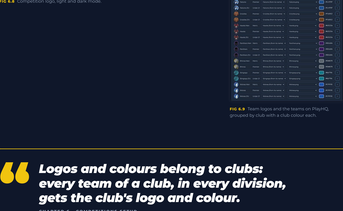


FIG 6.8 Competition logo, light and dark mode.

FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.

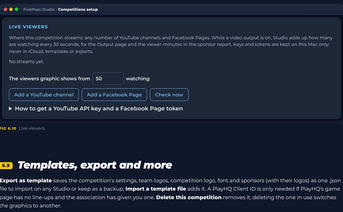


FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.

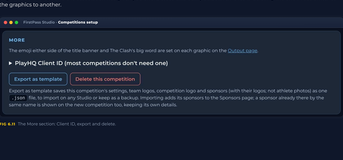


FIG 6.11 The More section: Client ID, export and delete.

CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers · 2 A sponsor · 3 Spots and moments · 4 Proof of play

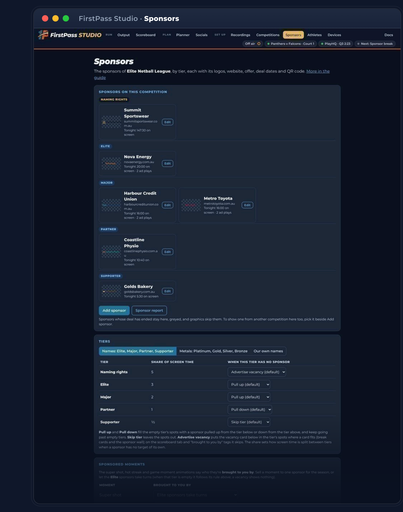


FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> **The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.**
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

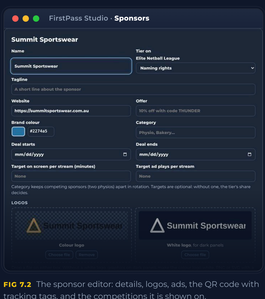


FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

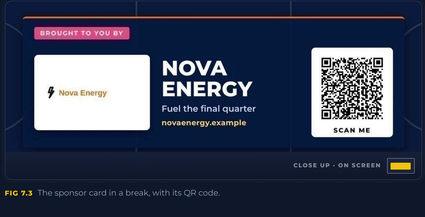


FIG 7.3 The sponsor card in a break, with its QR code.

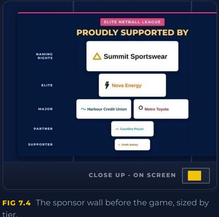


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play

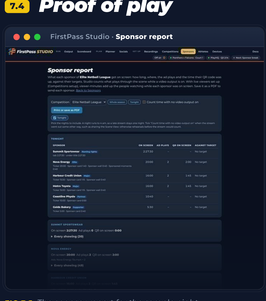


FIG 7.5 The sponsor report for the sample night.

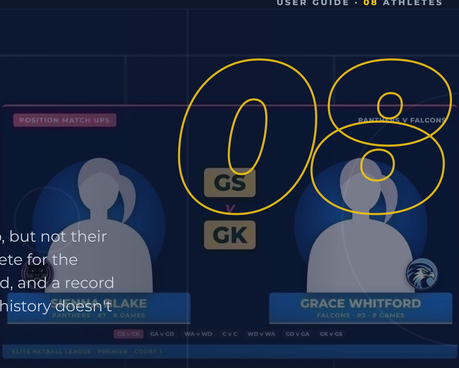
While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1. Photos
2. Positions and games played



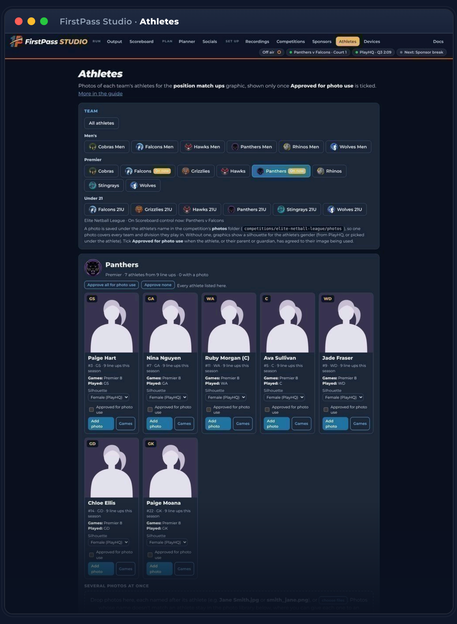


FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

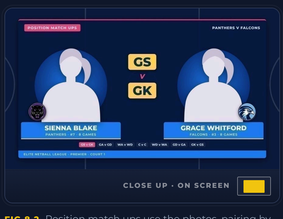


FIG 8.2 Position match ups use the photos, pairing by pairing.

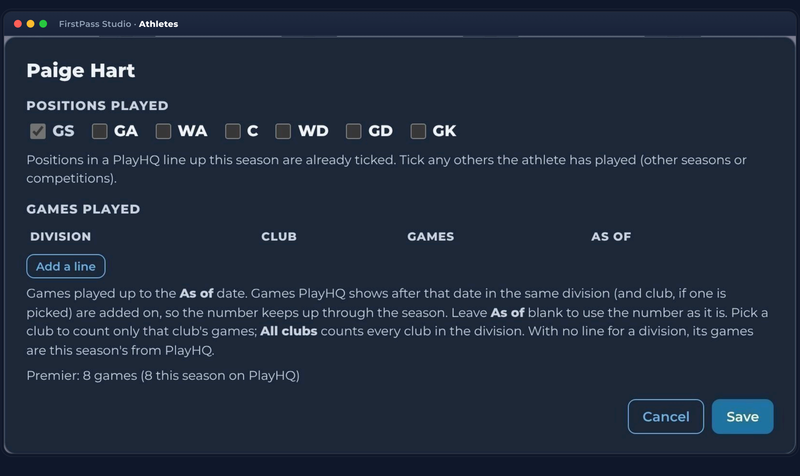


FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***
>
> CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link. It all lives on the **Devices** tab, under the list of devices allowed to change things (Stream Decks and their link sets, paired phones and programs), which is also where **Devices** in the menu bar icon goes.

1 The plugin · 2 Links for everything else · 3 Timed keys and the icon pack · 4 The Remote on a phone

FIG 9.1

The Devices tab: the devices allowed to change things at the top, then the plugin, then Links: every link Studio answers, each with what it does, Copy and its settings.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Each kind of key has its own colour and picture, so you can see what a key does at a glance. The numbered disc in the top right corner shows which output it acts on, in that output's colour.

KEY	LOOKS	PRESS	SHOWS
Source	Teal, an eye (pink with a speaker for video sound)	Show or hide (default), fade in, fade out; refresh the page or restart the video, video sound on or off	The source's name; lit while it's on, however it was turned on
Scene	Blue, stacked layers	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	Purple, a film strip	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**; the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Run order	Yellow, a numbered list	Play next (default), skip what's next, load a saved order, clear the list	What plays next, like **Play next ▶ / Quarter break / +2 more**; lit while something is queued, **empty** when nothing is; a Load key names its order and how many items it holds
Switcher	Red, a camera; no disc	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key. Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air, a yellow dot while the switcher isn't answering
Replay	Grey, a replay arrow; no disc	Play or pause, next or previous moment, nudge, stop the 4K replay or practice game running	What it acts on (**4K replay**, **Practice**, **Demo**); lit while one runs

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). Its disc shows the output's number in its colour (Output 1 orange, 2 pink, 3 blue, 4 green, as on the Output page), so two keys for the same source on different outputs can't be mixed up. A key for a removed output loses its disc and says **no output**.

> **Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.**
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Devices tab copy them, with this Mac's Stream Deck key after the port (`…8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Devices tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices can take that one away without touching your other keys. The Devices tab lists every one under **Links**, with what it does, the settings it takes and **Copy**. **Find** narrows the list, a graphic has **Preview** and a status read has **Try**. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Run order	Play what's next, skip it, clear the list, add a sequence or scene, load a saved order.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.



FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

9.4 The Remote on a phone

No Stream Deck to hand, or away from the desk? The **Remote** puts the status board and a Stream Deck's sequence keys on a phone. On the Mac, press **Open on your phone** at the top of the Socials tab, pick **Remote** and scan the QR code with the phone's camera, on the same Wi-Fi; the first time, choose **Allow everything** on the Mac when it asks, then add the page to the Home Screen to open it like an app (it's the Studio link with `/remote.html` on the end). At the top it says what's on at a glance: on air or off air and which outputs, the game's score, clock and whether PlayHQ is scoring it, the sequence playing and any about to start. Under that, **Play next ▶** and **Skip** for the run order, **Next step**, **Pause** and **Stop** for the sequence playing, and a key for every sequence in its colour, lit while it plays: press it to play the sequence on Output 1, press it again to stop it (which does its Then, as on the Output page). While the scoreboard is on **Manual** (a score press on the Mac or a Stream Deck switches it), a **Scoring by hand** card comes up with each team's score and +1, +2 (where super shots are worth 2) and −1 keys, the clock's Start and Stop and Next quarter; it goes away once the scoreboard follows PlayHQ again. A phone paired to score only can use those keys too.



CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu · 2 The data folder · 3 Settings in iCloud Drive · 4 Backup and restore · 5 Updates · 6 Links and other devices · 7 When something isn't right · 8 Data and bandwidth

10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices**; **User guide**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up...** and **Restore...**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Appearance** (Follow the Mac, Light, Dark), **Only this Mac can change things**, **What's new...** and Quit.

This guide comes with Studio: **User guide** in the menu bar icon, **Help > FirstPass Studio User Guide** while a Studio window is in front open it in a window of its own, and it is the **Docs** tab (⌘0) in any Studio window, with no internet needed. Phones and other computers on the network can read it too, at the Studio link with `/guide/` on the end.

10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 Settings in iCloud Drive

Settings folder > Move to iCloud Drive... moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac...** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac...** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 Backup and restore

Back up... saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore...** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

> **Updates download by themselves and install when the app quits, never mid-stream.**
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 Links and other devices

Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

Secure phone link. A phone's browser lets a page use its microphone (a phone as a backup commentator headset) or its share sheet only over https, which the `.local` link isn't. Turned on in the **Secure phone link** card on the Devices tab, Studio also answers at a name of its own on the internet, such as `https://k3f9x2.studio.firstpass.pivt.au:8443`, with a free Let's Encrypt certificate it renews by itself, and every **Open on your phone** QR code uses it; a phone pairs once more on the new link. The graphics, the Stream Deck and the http links carry on the same. It needs the internet at the venue, and a few routers refuse a name on the internet that points at their own network: then the https link doesn't open there and the http one still does. It's set up once for PIVT, then each Studio Mac is given the same three things:

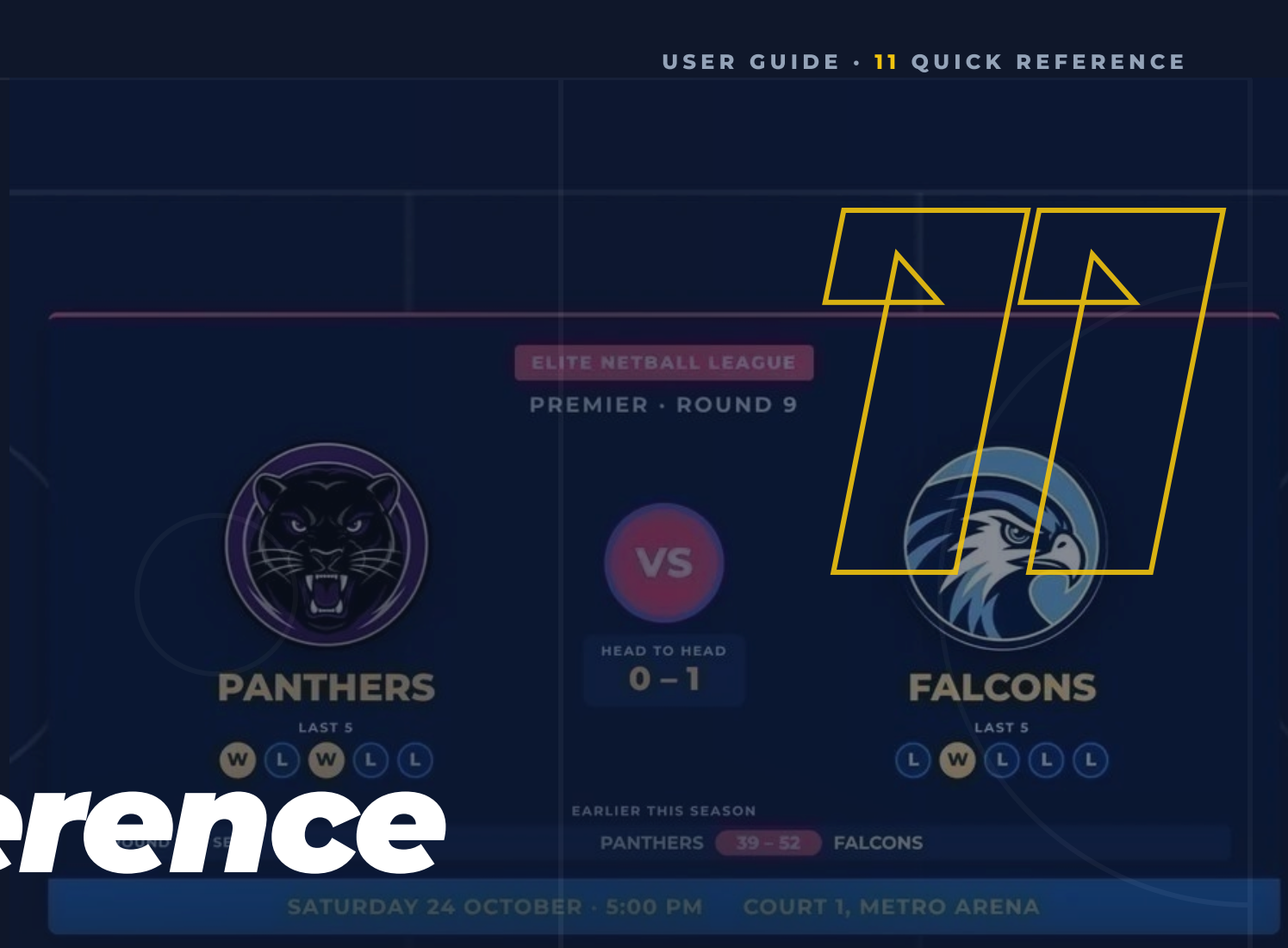
1. In AWS, **Route 53 › Hosted zones › Create hosted zone**: domain `studio.firstpass.pivt.au`, Public. Note its **Hosted zone ID** (starts with Z) and its 4 NS values.
2. In GoDaddy's DNS for **pivt.au**, add 4 records: type **NS**, name `studio.firstpass`, one NS value each. It can take up to an hour to work.
3. In **IAM**, a policy `FirstPassStudioDNS` allowing `route53:ChangeResourceRecordSets` and `route53:ListResourceRecordSets` on that hosted zone only, and `route53:GetChange`; a user `firstpass-studio-dns` with it attached and no console access; then **Security credentials › Create access key** (Application running outside AWS). Keep the secret somewhere safe: AWS shows it once.
4. On each Studio Mac, paste the hosted zone ID, the access key ID and the secret into the card and press **Turn on**. It takes about a minute the first time, and the card says each step and, if one fails, why and what to check. The secret goes in the Mac's Keychain.

10.7 When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering, or (PlayHQ chip purple, **PlayHQ not live**) the scorebench is off line. Fix the scorebench, or switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth. The PlayHQ line above the score on the Output page's status board shows it live: the ping, the bandwidth over the last minute and the data downloaded since Studio went on air.



CHAPTER 11 OF 11

Quick reference

1 Keyboard
2 Stream night in nine lines
3 Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01
Open Studio from the 1P menu and keep the Output page open all night.

02
Check the competition in use and the court on Competitions setup.

03
On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04
Turn on the video output your switcher takes and show one graphic to see it land.

05
Play a few seconds of music and an animation with sound.

06
Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07
Go to the Live game scene and let the sequences run the breaks.

08
Pick finals winners as they happen; turn on lower thirds when people appear.

09
Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links, and the devices allowed to change things	Devices
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

FirstPass STUDIO

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